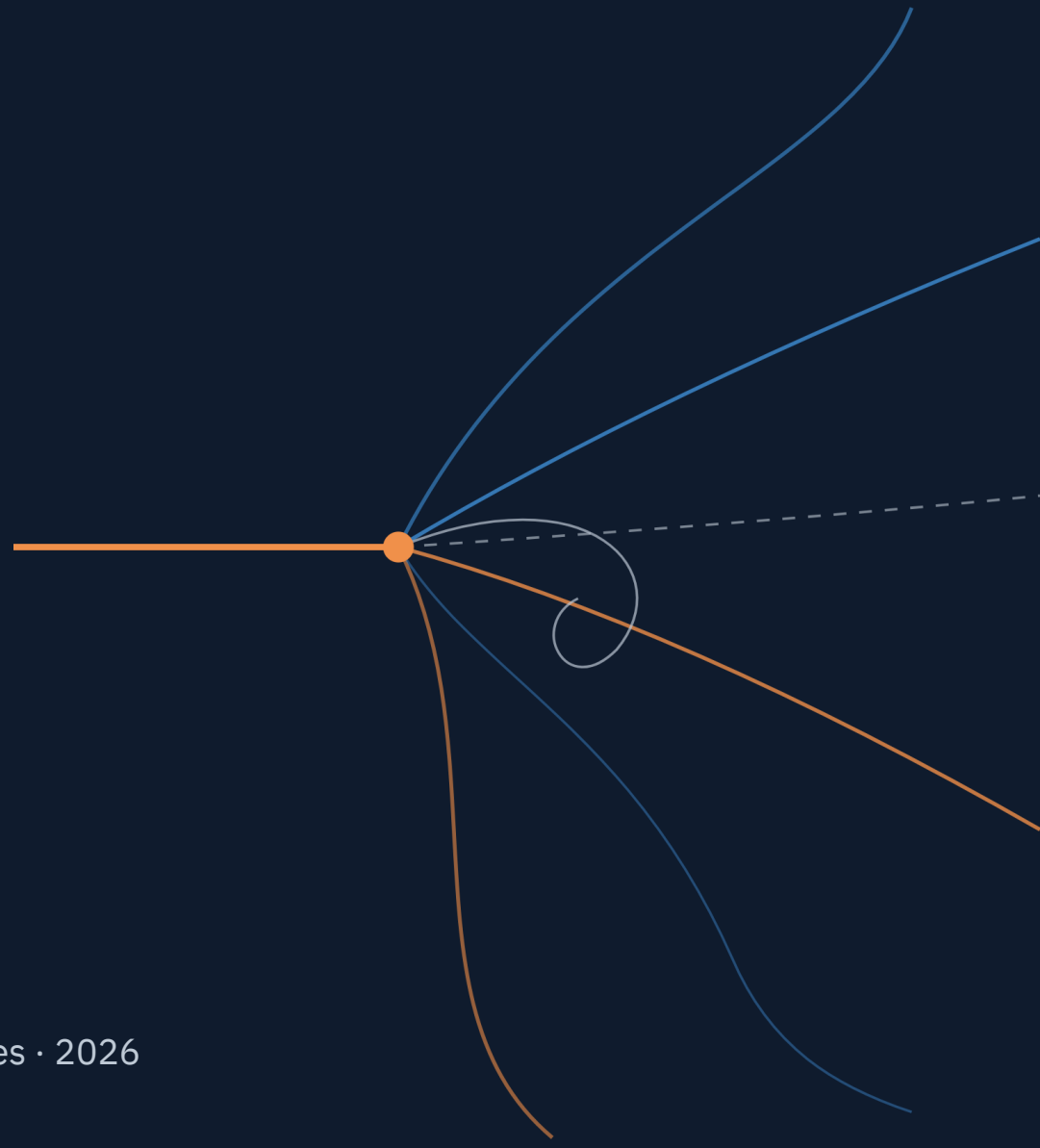
PARTICLE DETECTOR DESIGN AND SIMULATION

Radiation and Detectors

Radioactive sources, radiation–matter interactions and the general characteristics of detectors

Source: W. R. Leo, *Techniques for Nuclear and Particle Physics Experiments* — chapters 1, 2 and 5



Kaan Yüksel OYULMAZ · Ankara University Institute of Accelerator Technologies · 2026

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> **Source:** W. R. Leo, Techniques for Nuclear and Particle Physics Experiments, 2nd ed., Springer (1994).

OVERVIEW

From source to signal

1

Source

Radioactive nuclei, reactors, accelerators and cosmic rays emit α, β, γ and high-energy particles.

Leo · Ch. 1

2

Interaction

Radiation loses energy in matter by ionization, excitation, photon processes and nuclear reactions.

Leo · Ch. 2

3

Detector

The deposited energy becomes charge or light, then an electrical signal we can count and measure.

Leo · Ch. 5

To design a detector we must know what the source emits and how that radiation interacts with the detector material.

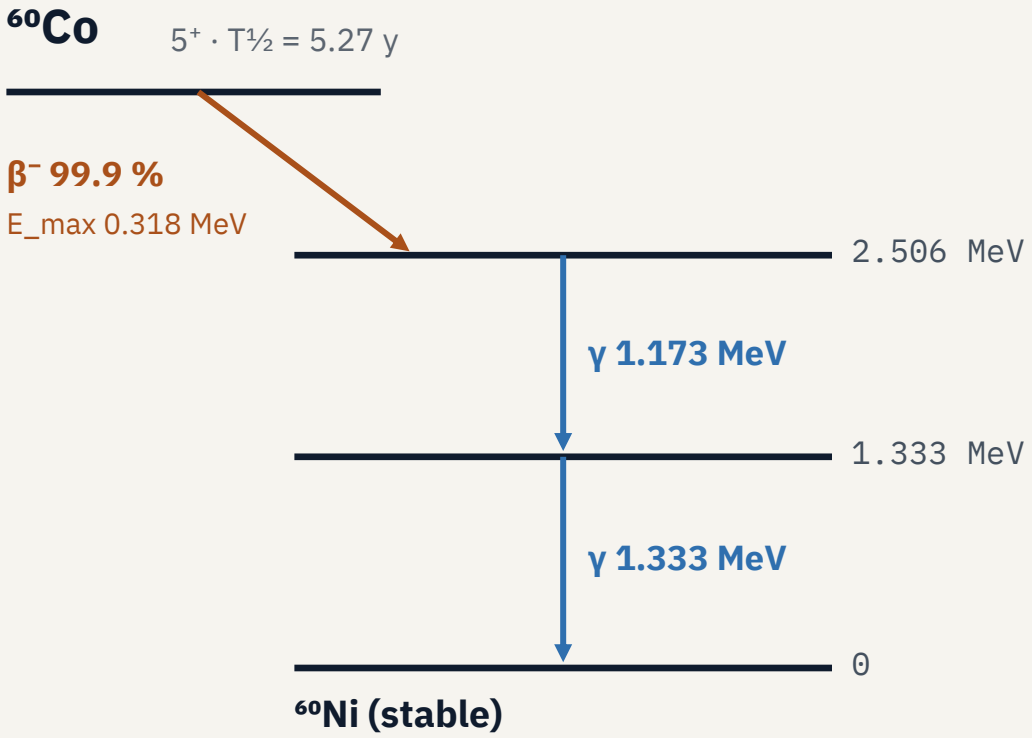
PART 1 · RADIOACTIVE SOURCES

Radioactive sources are the cheapest particle beams in the lab, and each decay mode emits a different kind of radiation

α	β^- β^+	EC	γ	e^+e^-
Alpha decay	**Beta decay**	**Electron capture**	**Gamma emission**	**Annihilation**
Monoenergetic He nuclei, 4–6 MeV	Continuous e^-/e^+ spectrum plus a neutrino	Followed by X-rays and Auger electrons	Discrete lines from excited nuclear states	Back-to-back pair of 511 keV photons

Key idea: a source is characterized by what it emits, at which energies, and how often (its activity).

Reading a nuclear level diagram



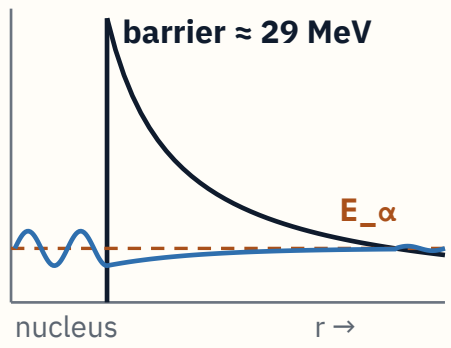
Two γ rays in cascade in every decay: a standard calibration source.

- Horizontal line: a nuclear energy level.
- Slanted arrow: a decay that changes Z (β⁻ to the right; β⁺, EC, α to the left).
- Vertical arrow: a γ transition within the same nucleus.
- Percentage: the fraction of decays taking that path (branching ratio).

Alpha decay

$(Z, A) \rightarrow (Z-2, A-4) + \alpha$

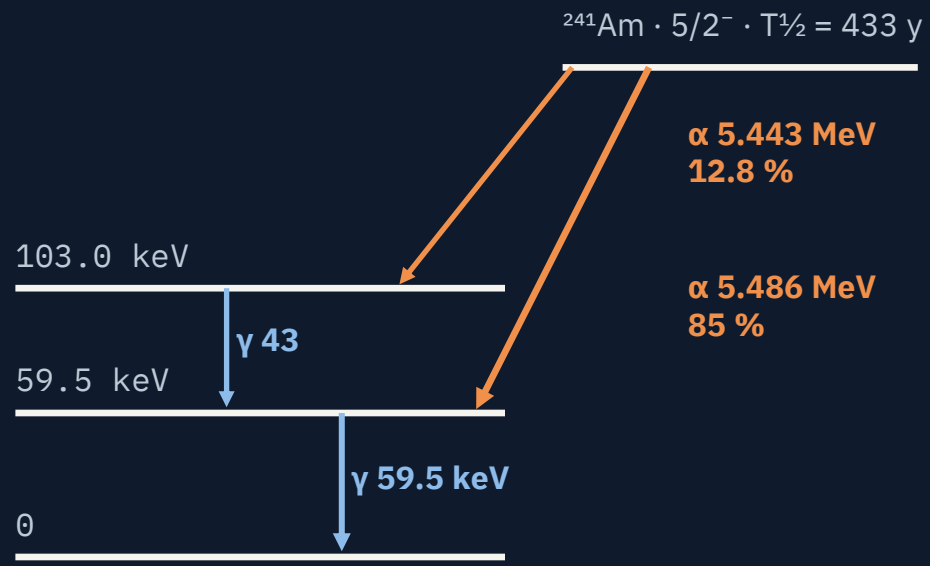
- Two-body decay → monoenergetic lines: $T_\alpha \approx Q \cdot (A-4)/A$
- Energies lie between 4 and 6 MeV.
- Range in air is a few cm, so sources are made very thin.



What is tunnelling?

The α energy (≈ 5.5 MeV) is far below the Coulomb barrier (≈ 29 MeV): classically it cannot escape. In quantum mechanics the wave function leaks through the barrier, decaying inside it → a small escape probability.

Example: ^{241}Am decay scheme



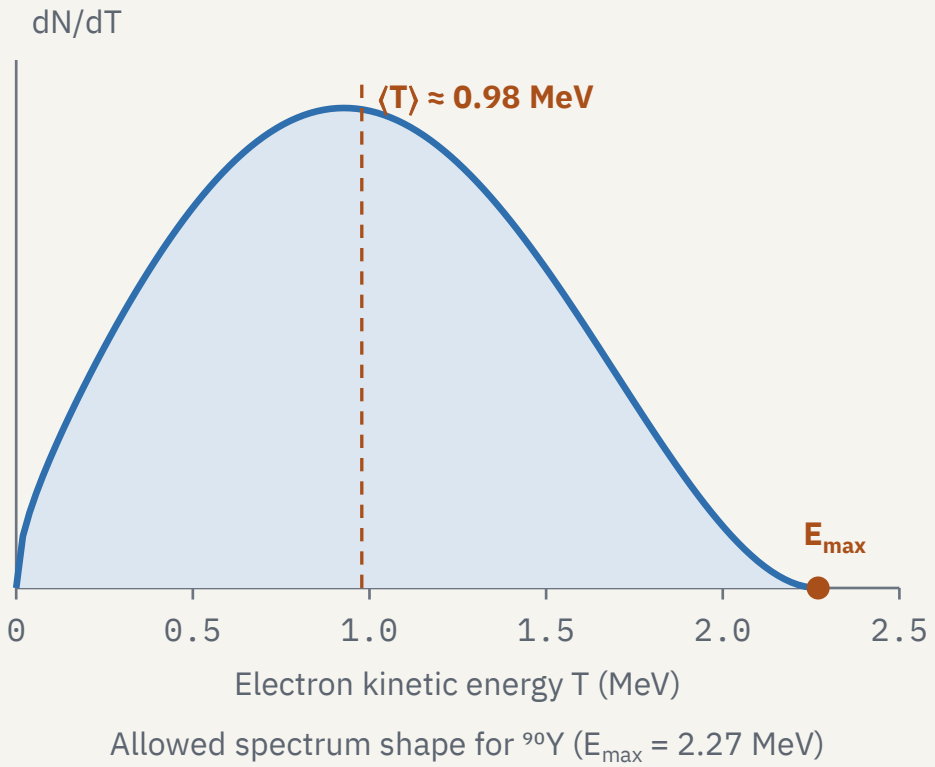
Not to scale · Q_α = 5.64 MeV · ratios: Leo, Table 1.2

Beta decay: a continuous spectrum

β^- : $n \rightarrow p + e^- + \bar{\nu}_e$

β^+ : $p \rightarrow n + e^+ + \nu_e$

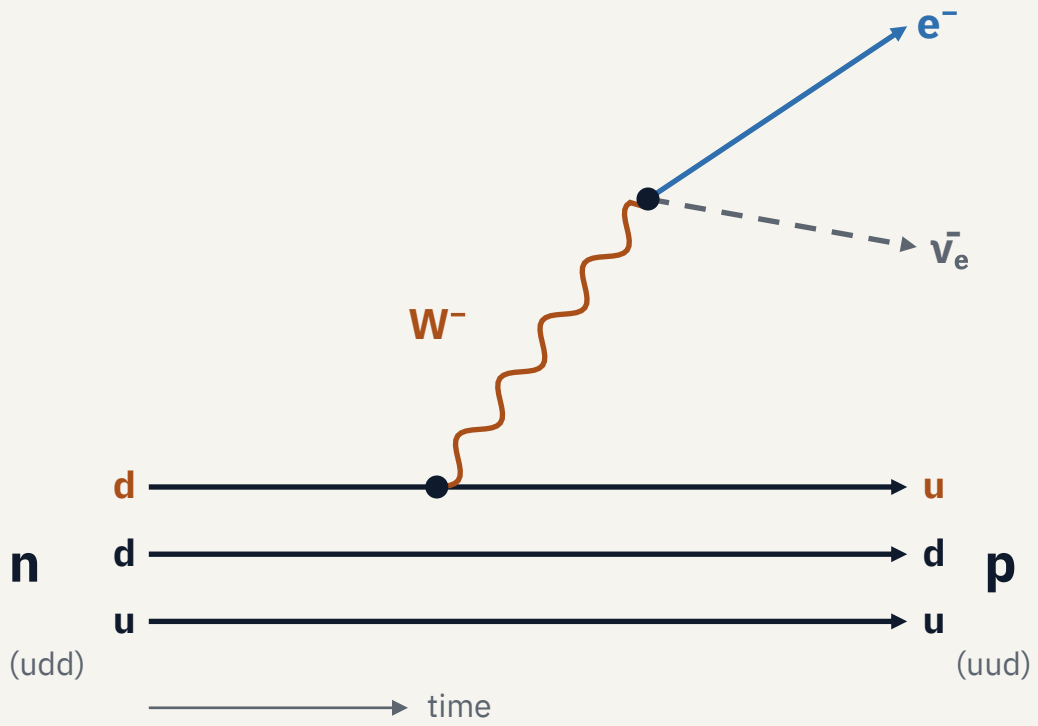
- The energy is shared with an undetected (anti)neutrino → a continuous spectrum from 0 to E_{max} (tens of keV to a few MeV).
- Most β sources leave the daughter nucleus excited and also emit γ rays.
- Pure β^- emitters are rare: ^{3}H (18.6 keV), ^{14}C (156 keV), ^{90}Sr/^{90}Y (0.546 / 2.27 MeV).
- Sources must be thin; otherwise a thick β^+ source gives a large 511 keV background.



Allowed spectrum shape for ^{90}Y (E_{max} = 2.27 MeV)

Beta decay

$$d \rightarrow u + W^- \rightarrow u + e^- + \bar{\nu}_e$$



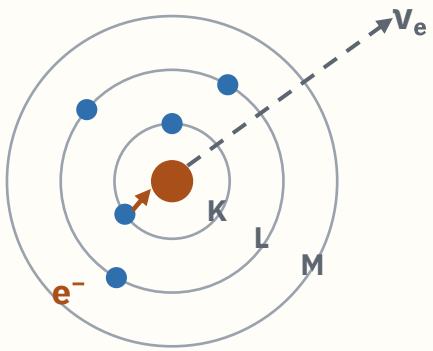
- A d quark of the neutron (udd) emits a W^- boson and turns into a u quark → proton (uud).
- The W^- immediately decays into an electron and an electron antineutrino.
- Conservation: charge 0 = (+1) + (−1) + 0; lepton number 0 = 0 + 1 − 1.
- β^+ (inside a nucleus): u → d + W^+, $W^+ \rightarrow e^+ + \nu_e$.

Why the weak interaction?

W mass ≈ 80 GeV: the interaction is very short-ranged and feeble, so half-lives are long (free neutron: $T_{1/2} \approx 10$ min).

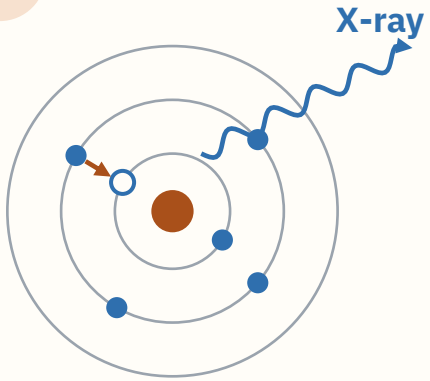
Electron capture and Auger electrons

1 Capture



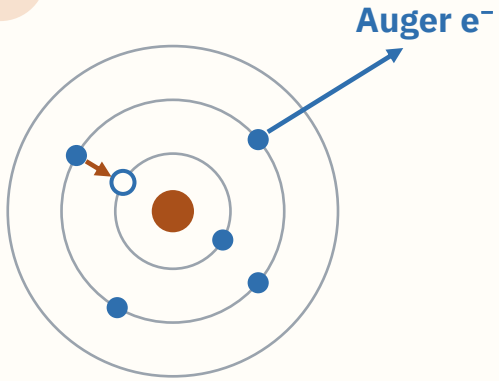
The nucleus captures a K electron: $p + e^- \rightarrow n + \nu_e$ (an alternative to β^+). A vacancy is left in the K shell.

2a X-ray



An electron from the L shell fills the K vacancy; the energy difference comes out as a characteristic X-ray.

2b Auger electron

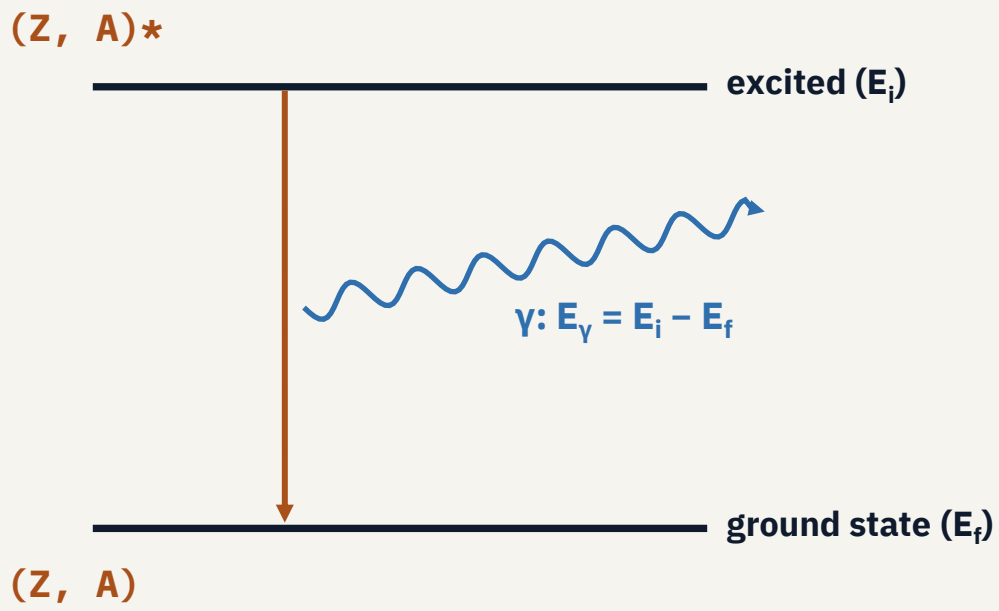


The same energy is given to another electron, which is ejected from the atom (Auger e^-).

X-ray or Auger? Mostly Auger in light elements ($\lesssim$ a few keV, easily absorbed), X-rays in heavy ones.

Example ^{55}Fe → ^{55}Mn: only 5.9 keV Mn Kα X-rays, no γ ($T_{1/2}$ = 2.74 y).

Gamma decay

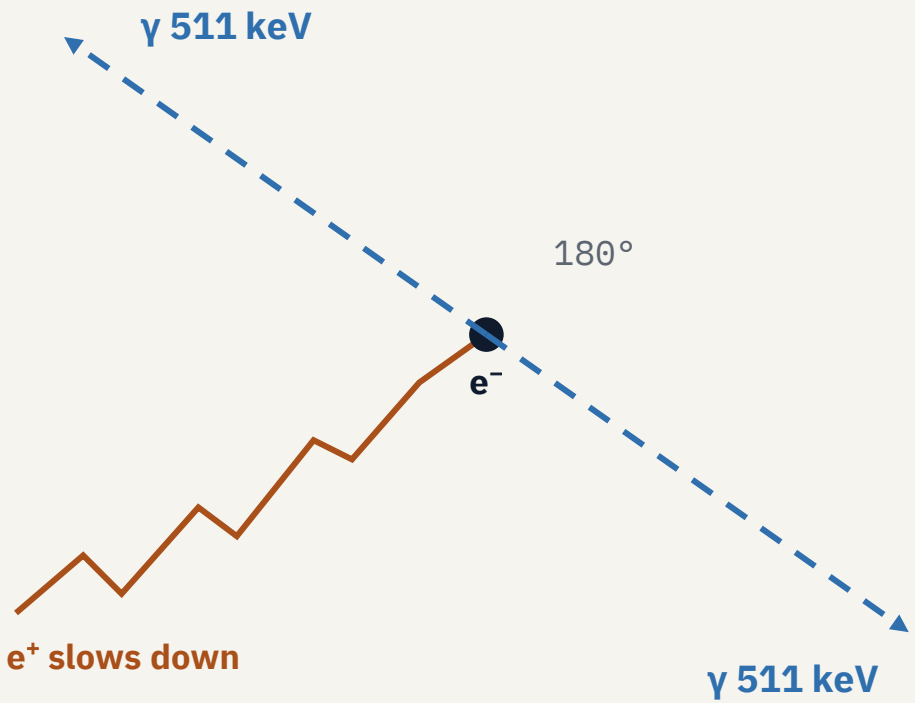


$(Z, A)^* \rightarrow (Z, A) + \gamma$

- An excited nucleus drops to a lower level and emits the difference as a γ photon; Z and A do not change.
- E_γ = level difference: a few hundred keV to a few MeV
- The excited state usually comes from a preceding α or β decay (^{60}Co).
- γ rays come from the nucleus, X-rays from the electron shell.

Source	γ energy	Note
^{137}Cs	662 keV	most common calibration
^{60}Co	1173 + 1333 keV	two γ in cascade
^{241}Am	59.5 keV	low energy

Annihilation radiation



- The positron slows down in matter, then annihilates with an electron: $e^+ + e^- \rightarrow \gamma + \gamma$
- At rest: two photons emitted back to back, each with $m_e c^2$ = 511 keV.
- Every β^+ source is also a 511 keV γ source.

Example: ^{22}Na (T½ = 2.60 y)

511 keV	1275 keV
annihilation, ×2	nuclear γ

Every detector relies on radiation losing energy in its material: how does each particle do it?

p α ions

Heavy charged particles

Many small collisions with atomic electrons; straight tracks, well-defined range.

e^- e^+

Electrons and positrons

Collisions and bremsstrahlung; zigzag paths, strong multiple scattering.

γ X-Ray

Photons

All or nothing: photoelectric, Compton, pair production; exponential attenuation.

> **Charged vs neutral:** charged particles lose energy continuously (dE/dx); neutral ones interact rarely but drastically (cross section σ).

Cross section and mean free path

1. $N = \rho \cdot N_A / A$

   Target density: atoms per cm³ (ρ: g/cm³, A: g/mol).

2. $P(dx) = N\,\sigma\,dx$

   Probability of interacting in a thin layer dx; σ: cross section (1 b = 10^{-24} cm²).

3. $P(x) = e^{-N\sigma x}$

   Probability of crossing a thickness x without any interaction: exponential decrease.

4. $\mu = N\sigma, \quad \lambda = 1/\mu$

   Attenuation coefficient μ and mean free path λ.

> **Areal density ρx (g/cm²):** 1 cm of water = 1 g/cm², 1 cm of Pb = 11.35 g/cm².

Example: 662 keV γ in lead (ρ = 11.35 g/cm³)

1) $N = 11.35 \times 6.022 \cdot 10^{23} / 207.2$
   $= 3.30 \cdot 10^{22}$ atoms/cm³
2) $\sigma \approx 39$ b $= 3.9 \cdot 10^{-23}$ cm² (NIST)
3) $\mu = N\sigma = 1.29$ cm^{-1} → $\lambda = 0.78$ cm
4) $P(1\text{ cm}) = e^{-1.29} = 0.28$

28 % of the photons cross 1 cm of lead without interacting, 72 % interact.

Geant4 link: the sampling $s = -\lambda \ln \xi$ is exactly this exponential law.

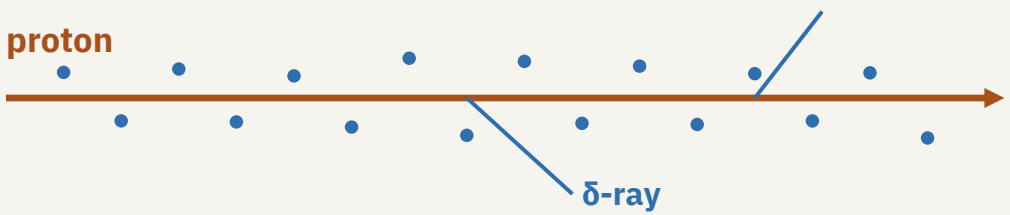
Heavy charged particles in matter

Collisions with atomic electrons

Dominant process: soft collisions excite, hard collisions ionize, and each one transfers only a small fraction of the energy.

Elastic scattering off nuclei

Rarer: transfers very little energy (heavy nuclei), mainly deflects the particle.



Maximum energy transfer

$$W_{max} \approx 2m_e c^2 \beta^2\gamma^2 \qquad (M \gg m_e)$$

A head-on collision with a light electron takes only a small piece of the particle's energy.

Example: 10 MeV proton

$$\beta\gamma = 0.146 \rightarrow W_{max} \approx 22\ \text{keV}$$

but there are so many collisions that it stops in only 0.25 mm of copper along an almost straight track (Leo).

The Bethe–Bloch formula

$$-\frac{dE}{dx} = K\,\rho\,\frac{Z}{A}\,\frac{z^2}{\beta^2} \times \left[\ln\left(\frac{2m_e c^2\beta^2\gamma^2 W_{max}}{I^2}\right) - 2\beta^2 - \delta - \frac{2C}{Z}\right]$$

$$K = 2\pi N_a r_e^2 m_e c^2 = 0.1535\ \text{MeV cm}^2/\text{g}$$

As the charged particle passes, it gives each electron a small Coulomb kick; dE/dx is the sum of these kicks.

$K\,\rho\,Z/A$

Electron density: the more electrons the material contains, the larger the loss.

z^2

Particle charge: at the same speed an α (z = 2) loses 4 times more than a proton.

$1/\beta^2$

A slow particle stays longer near each electron, so the kick and the loss grow.

$\ln(\ldots/I^2)$

The range between the largest (W_{max}) and smallest (~I) transfer; varies slowly.

I: mean excitation potential (water: 75 eV) · δ: density correction (high E) · C: shell correction (low E) · $W_{max} \approx 2m_e c^2\beta^2\gamma^2$

Bethe–Bloch: a worked example

Data: proton, T = 10 MeV, Mc^2 = 938.3 MeV, z = 1 · water: ρ = 1 g/cm³, Z/A = 0.555, I = 75 eV

1 Kinematics

$\gamma = 1 + T/Mc^2 = 1 + 10/938.3 = 1.0107$

$\beta^2 = 1 - 1/\gamma^2 = 0.0210, \quad \beta^2\gamma^2 = 0.0214$

2 Maximum transfer

$W_{max} \approx 2m_e c^2\beta^2\gamma^2 = 1.022 \times 0.0214 = 0.0219$ MeV

3 Logarithm

$\ln(2m_e c^2\beta^2\gamma^2 \cdot W_{max} / I^2) = \ln(8.5 \cdot 10^4) = 11.35$

$[\ \dots\ ] = 11.35 - 2\beta^2 = 11.31 \quad (\delta, C \approx 0)$

4 Prefactor

$K\ \rho\ (Z/A)\ z^2/\beta^2 = 0.1535 \times 1 \times 0.555 / 0.0210 = 4.06$ MeV/cm

5 Result

$-dE/dx = 4.06 \times 11.31 \approx 45.9$ MeV/cm

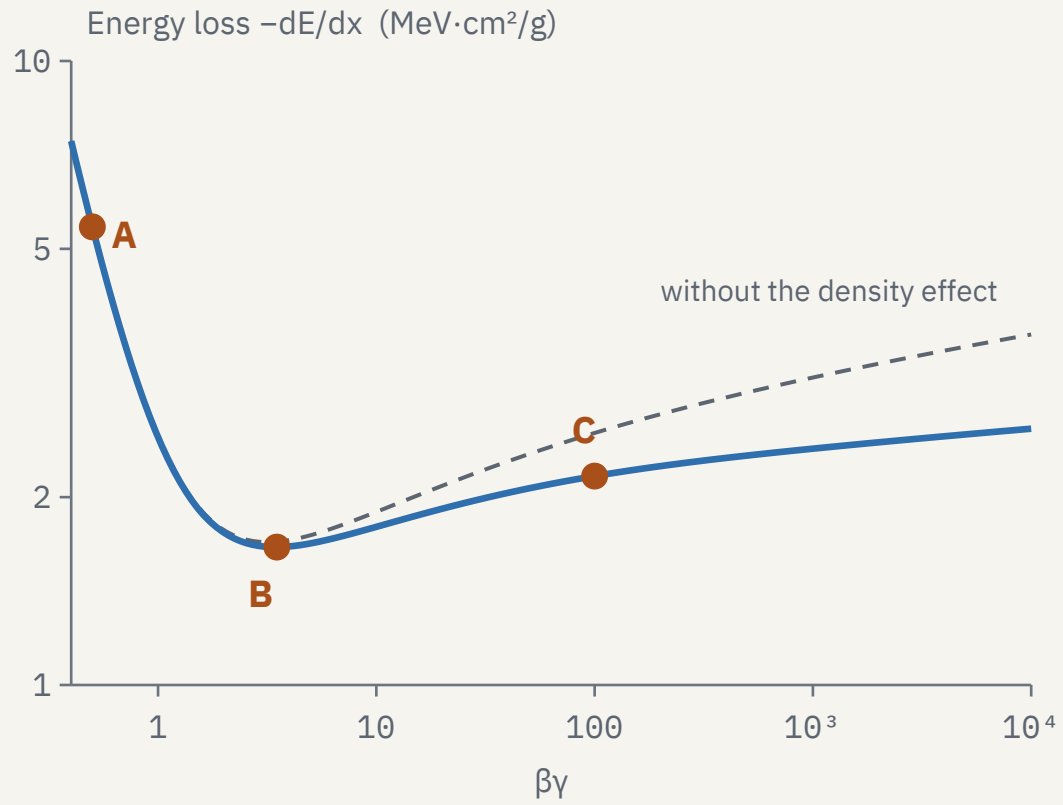
Result

45.9 MeV/cm

NIST PSTAR: 45.7 MeV/cm

- A 10 MeV proton travels only ≈ 1.2 mm in water.
- The proton can give an electron at most ≈ 22 keV → thousands of collisions.
- An α at the same speed (z = 2): ≈ 4 × 45.9 ≈ 184 MeV/cm.

How does energy loss depend on speed?



Muon in silicon, computed with the Bethe–Bloch formula

Vertical axis: energy lost per cm of path, divided by the material density (MeV·cm²/g). Multiply by the density (g/cm³) to get MeV/cm.

Horizontal axis: βγ = p/(Mc); β = v/c is the speed, γ the Lorentz factor. At βγ = 1 the speed is ≈ 0.71c.

A **Slow particle (βγ < 1)**

Loss rises steeply (≈ 1/β²): a slow particle stays longer near each electron.

B **Minimum (βγ ≈ 3–4)**

Loss is smallest, ≈ 1–2 MeV·cm²/g. These are called minimum ionizing particles (MIPs).

C **Very fast particle (βγ ≫ 10)**

Loss rises very slowly; the density effect limits this rise.

Computing the energy loss: two examples

Example 1: muon in silicon (ρ = 2.33 g/cm³)

Point	βγ	Kinetic energy	MeV·cm²/g	MeV/cm
A	0.5	12.5 MeV	5.42	12.63
B	3.5	278.9 MeV	1.66	3.87
C	100	10.5 GeV	2.16	5.03

MeV/cm = (MeV·cm²/g) × ρ. For point B: 1.66 × 2.33 = 3.87 MeV/cm.

In a 300 µm (0.03 cm) silicon detector: 3.87 × 0.03 ≈ 0.116 MeV = 116 keV deposited.

> **A, B, C:** the points on the previous slide's plot. B (MIP) deposits the least energy.

Example 2 (Leo, Example 2.1)

Cosmic-ray muons cross a 2 cm thick plastic scintillator. How much energy do they deposit on average?

Cosmic muons are very energetic, i.e. minimum ionizing: in plastic dE/dx ≈ 1.9 MeV·cm²/g.

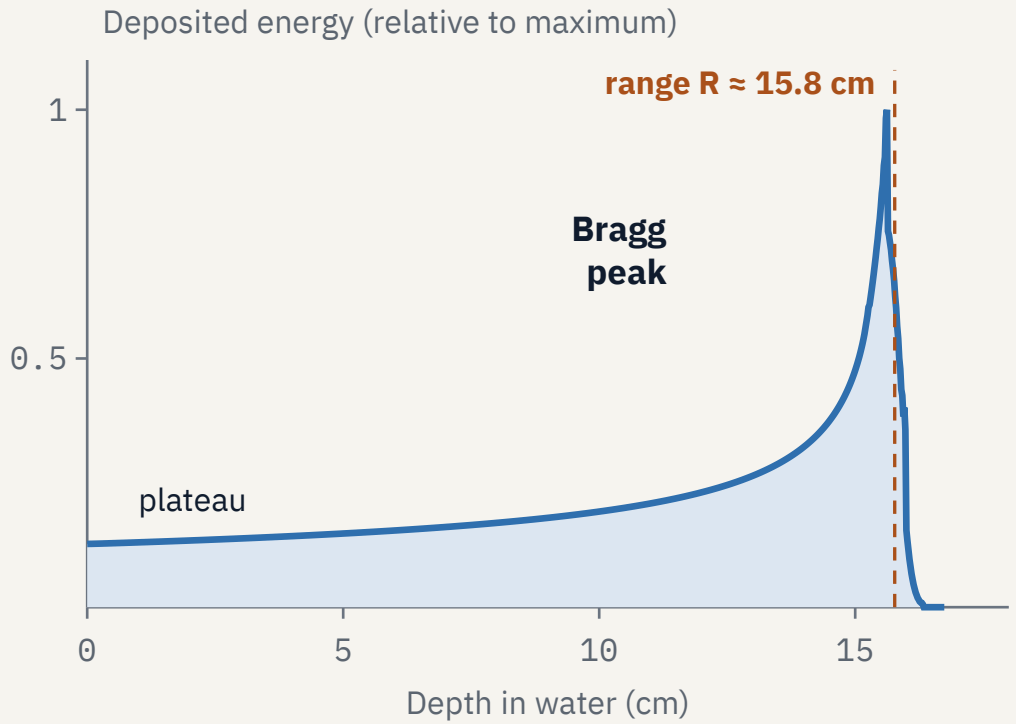
$$\Delta E = (dE/dx) \cdot \rho \cdot x$$

ΔE: deposited energy, ρ = 1.03 g/cm³: density of the plastic, x = 2 cm: thickness

$$\Delta E = 1.9 \times 1.03 \times 2 \approx 3.9 \text{ MeV}$$

They all deposit about the same energy: a peak forms in the spectrum and is used for calibration.

Bragg peak and range



Why a peak at the end? As the proton slows down it loses more energy; it deposits most of its energy just before stopping (150 MeV proton in water).

Range R: the path the particle travels in the material before it stops.

$$R = \Sigma\ \Delta x = \Sigma\ \Delta E\ /\ (dE/dx)$$

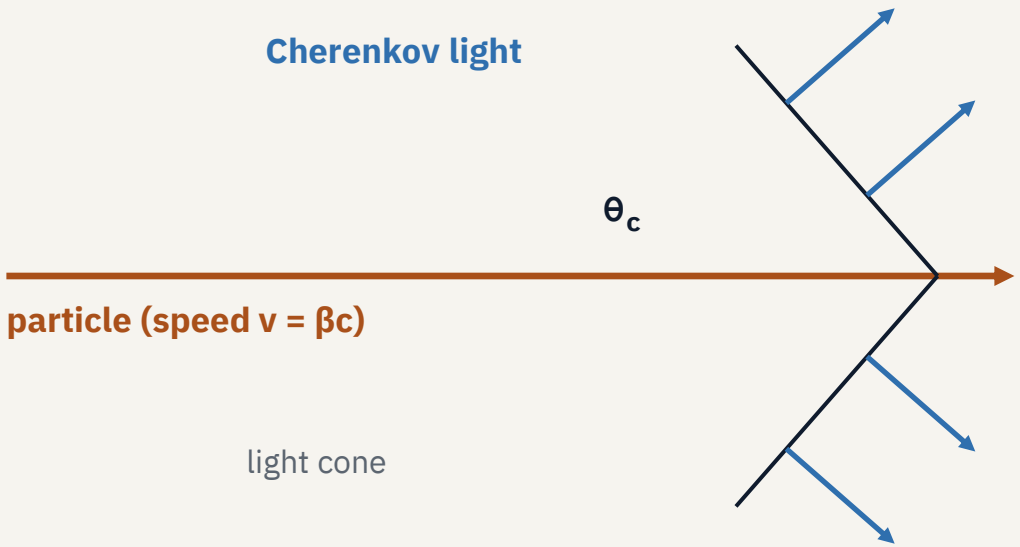
Split the energy into small slices ΔE. In each slice the path travelled is Δx = ΔE / (dE/dx). Add them all up.

Example: 10 MeV proton in water, ΔE = 2 MeV

Energy slice	dE/dx (mid-slice)	Δx = ΔE/(dE/dx)
10 – 8 MeV	50 MeV/cm	0.40 mm
8 – 6 MeV	61 MeV/cm	0.33 mm
6 – 4 MeV	80 MeV/cm	0.25 mm
4 – 2 MeV	120 MeV/cm	0.17 mm
2 – 0 MeV	270 MeV/cm	0.07 mm
Total		**1.22 mm**

NIST: 1.23 mm. The last 2 MeV are deposited in only 0.07 mm: the Bragg peak.

Cherenkov Radiation



What happens? Light travels at c/n in a medium. If the particle is faster than that, the light it emits cannot keep ahead and forms a cone, like the shock wave of a supersonic aircraft.

The same calculation gives a threshold energy of 55 MeV for a muon and 485 MeV for a proton.

n: refractive index = c / (speed of light in the medium). Water 1.33, glass ≈ 1.5, air 1.0003

β = v/c: the particle's speed in units of the speed of light

cos θ_c = 1/(βn): the angle between the light and the particle direction

Threshold: β > 1/n. A slower particle emits no light.

Example: when does an electron emit light in water?

1) Threshold speed: $\beta = 1/1.33 = 0.752$
2) Lorentz factor: $\gamma = 1/\sqrt{(1 - 0.752^2)} = 1.517$
3) Threshold energy: $T = (\gamma - 1)\cdot m_e c^2 = 0.26$ MeV
4) Very fast electron (β ≈ 1): $\theta_c = 41°$

$m_e c^2$ = 0.511 MeV: rest-mass energy of the electron

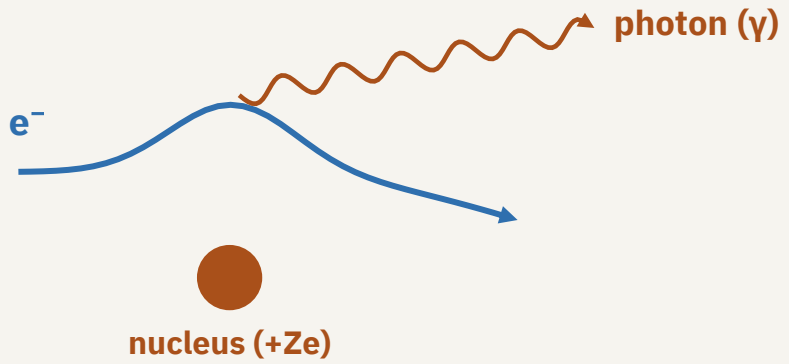
Electrons: two ways to lose energy

1 Collision (ionization)

Like protons, they lose energy by hitting atomic electrons. But the electron is very light, so it is easily deflected and can lose up to half its energy in a single collision.

2 Bremsstrahlung (braking radiation)

Passing a nucleus, the electron changes direction and emits a photon. This loss grows with energy; it is negligible for heavy particles (about 40 000 times smaller for a muon).



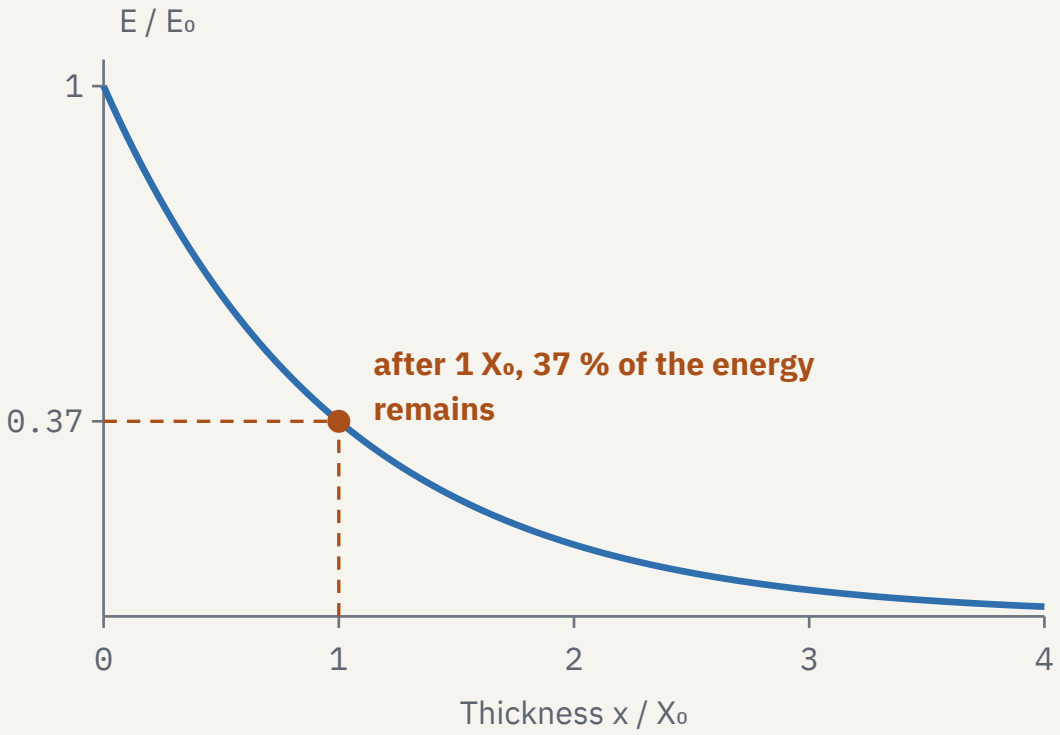
Critical energy E_c

The energy at which the two losses are equal. Below E_c collisions dominate, above it bremsstrahlung.

$$E_c \approx 800\ \text{MeV} / (Z + 1.2)$$

Z: atomic number of the material.
Lead (Z = 82): 800 / 83.2 ≈ 9.6 MeV.
Water: ≈ 92 MeV.

Radiation length X_0



X_0: the thickness over which a high-energy electron drops to 1/e (≈ 37 %) of its energy through bremsstrahlung.

$$E(x) = E_0 \cdot e^{-x/X_0}$$

E_0: initial energy · x: material thickness

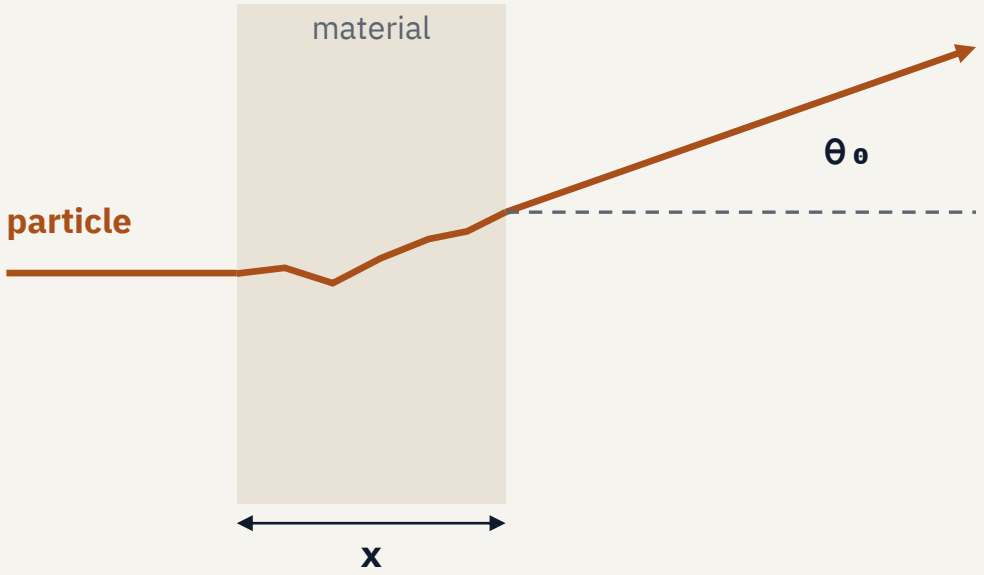
Material	X_0	Note
Air	300 m	very long
Water	36.1 cm	
Lead	0.56 cm	very short

Example: 1 GeV electron, 1 cm of lead

$x/X_0 = 1/0.56 = 1.79$

$E = 1000\ \text{MeV} \times e^{-1.79} \approx 167\ \text{MeV}$

Multiple Coulomb scattering



What happens? Passing each nucleus the particle is deflected very slightly. These small deflections add up randomly; the particle exits at a slightly different angle.

$$\theta_0 = \frac{13.6\ \text{MeV}}{\beta c p} \cdot z \cdot \sqrt{x/X_0} \cdot \left[1 + 0.038 \cdot \ln(x/X_0)\right]$$

θ_0: typical deflection angle (rad) · p: momentum (MeV/c)
β: speed (v/c) · **z:** charge (1 for proton, muon)
x: thickness · **X_0:** radiation length (previous slide)

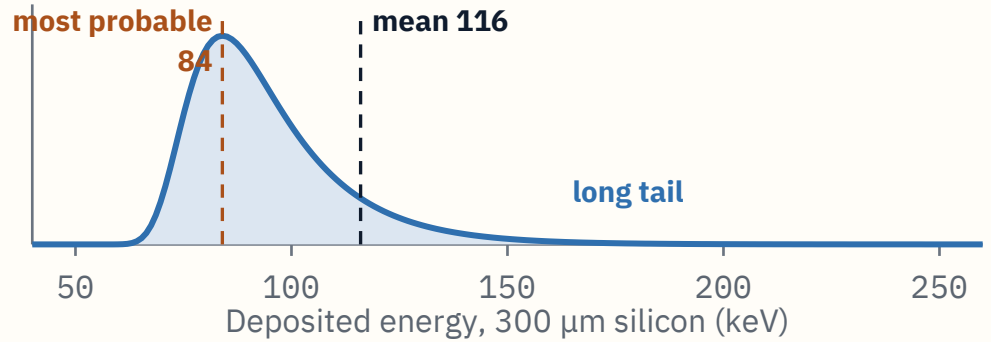
Example: 1 GeV/c muon, 300 µm of silicon (X_0 = 9.37 cm)

1) $x/X_0 = 0.03 / 9.37 = 0.0032$
2) $\sqrt{x/X_0} = 0.057$; $[\ \ldots\ ] = 0.78$
3) $\theta_0 = (13.6/1000) \times 0.057 \times 0.78$
   ≈ 0.0006 rad = 0.6 mrad
4) Shift after 1 cm: 0.6 mrad × 10 mm = 6 µm

Energy loss fluctuations

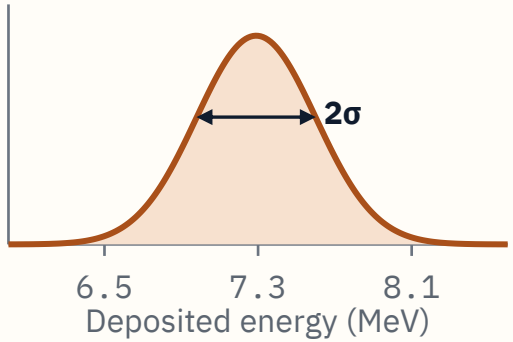
Bethe–Bloch gives the mean loss. But each particle makes a different number of collisions, so the deposited energy varies from particle to particle.

Thin detector: Landau distribution



Few collisions; an occasional large transfer (δ-ray) makes a long tail to the right. The most probable value is below the mean.

Thick absorber: Gaussian distribution



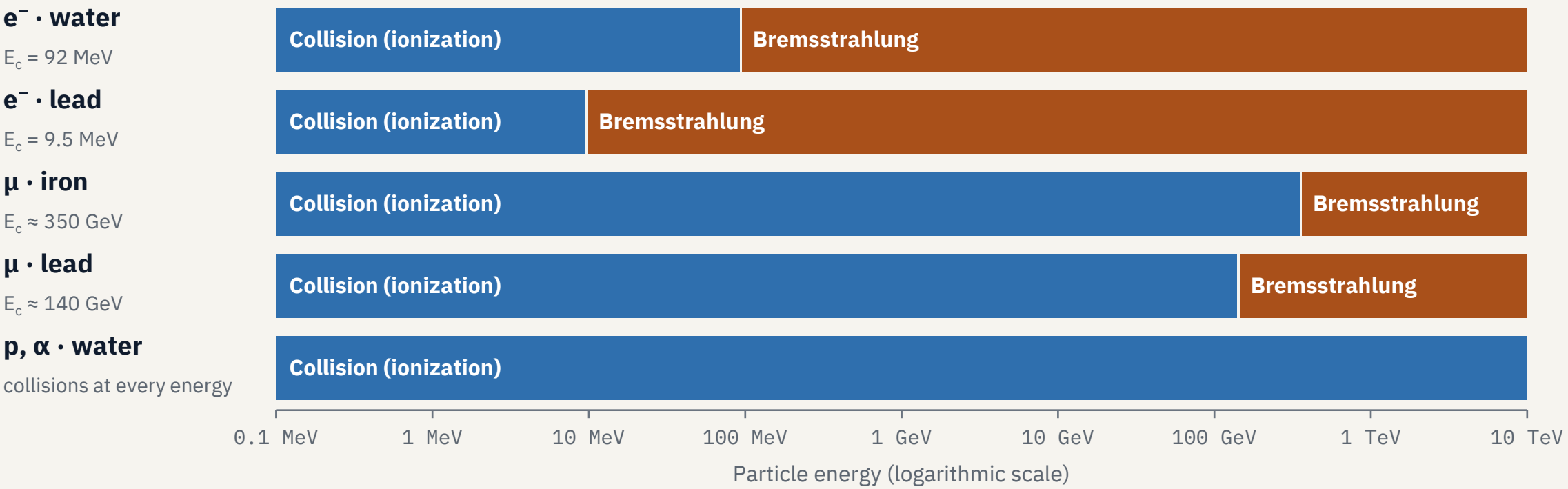
100 MeV p, 1 cm water

$$\sigma^2 = 0.1569 \cdot \rho \cdot (Z/A) \cdot x$$
$$= 0.1569 \times 1 \times 0.555 \times 1$$
$$= 0.087 \text{ MeV}^2$$
$$\sigma \approx 0.31 \text{ MeV}$$

The sum of many small contributions gives a symmetric Gaussian (Bohr). Result: loss 7.3 ± 0.3 MeV.

Which one? κ = mean loss / W_{max} (maximum transfer in a single collision). Small κ: Landau; $\kappa \gtrsim 1$: approaches a Gaussian ($\kappa \approx 2$ in the example).

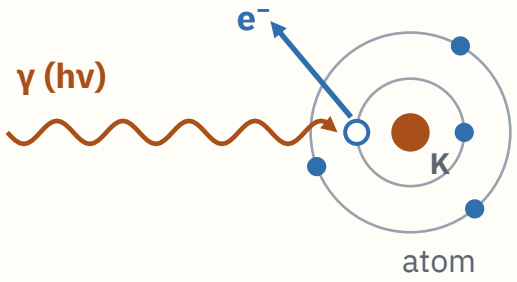
Which process dominates for charged particles?



> **The boundary is the critical energy E_c:** collisions dominate below it, bremsstrahlung above. Bremsstrahlung scales as $1/\text{mass}^2$; the muon is 207 times heavier than the electron, so it sets in much later.

Photons: three basic processes

Photoelectric effect

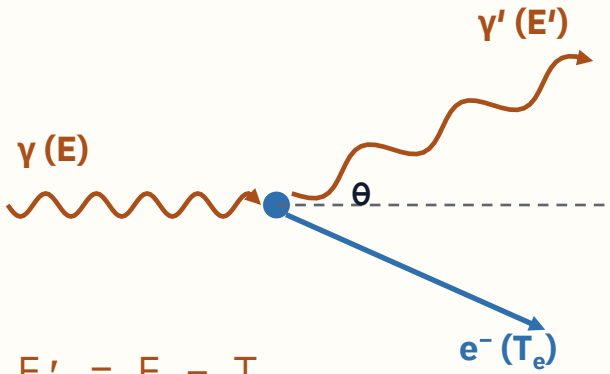


$$T_e = h\nu - B$$

The photon gives all its energy to a bound electron and disappears. The electron is ejected from the atom.

hν: photon energy · B: binding energy

Compton scattering

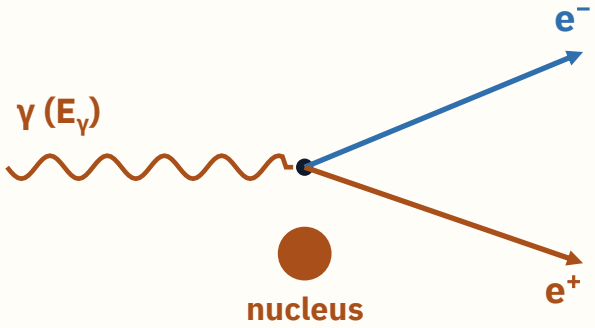


$$E' = E - T_e$$

The photon hits an electron, gives it part of its energy and changes direction.

T_e: energy taken by the electron

Pair production



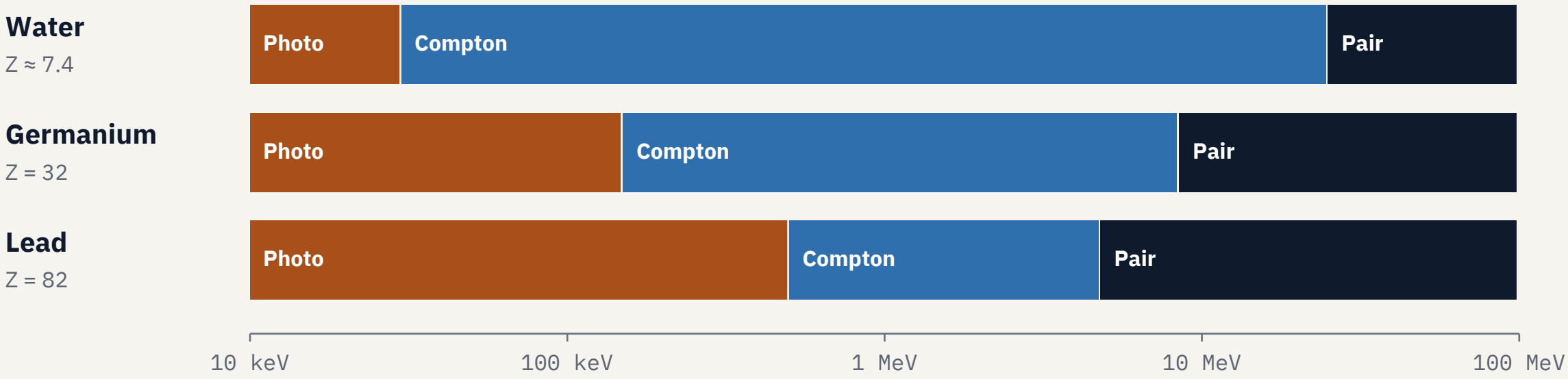
$$E_\gamma \geq 2m_e c^2 = 1.022\ \text{MeV}$$

Near a nucleus the photon turns into an electron and a positron.

$m_e c^2$ = 0.511 MeV (electron mass)

Photons have no range: a photon leaves the beam in a single interaction, so the beam is attenuated exponentially.

Which process dominates?



- The boundaries are where two cross sections are equal: as Z grows the photoelectric region widens and pair production starts earlier.
- γ spectrometers use high-Z materials (NaI, BGO, HPGe) to make full absorption likely.

LET'S APPLY WHAT WE LEARNED · GEANT4

A single-slab experiment with Geant4

Setup

Slab: 2 cm of aluminium (Z = 13), vacuum around it
Source: a single point 1 mm in front of the slab, normal to it
Particle: 10 000 photons
Energy: random between 1 keV and 100 MeV

What do we count?

Each photon's **first interaction** in the slab:
• photoelectric • Compton • pair production
• passed through without interacting

Rayleigh scattering is not counted: the photon keeps its energy.

What do we compute?

The energy axis is split into 25 bins; in each bin

$$\mu = -\ln(N_{pass} / N) / x$$
$$\mu_i = \mu \cdot n_i / n_{int}$$

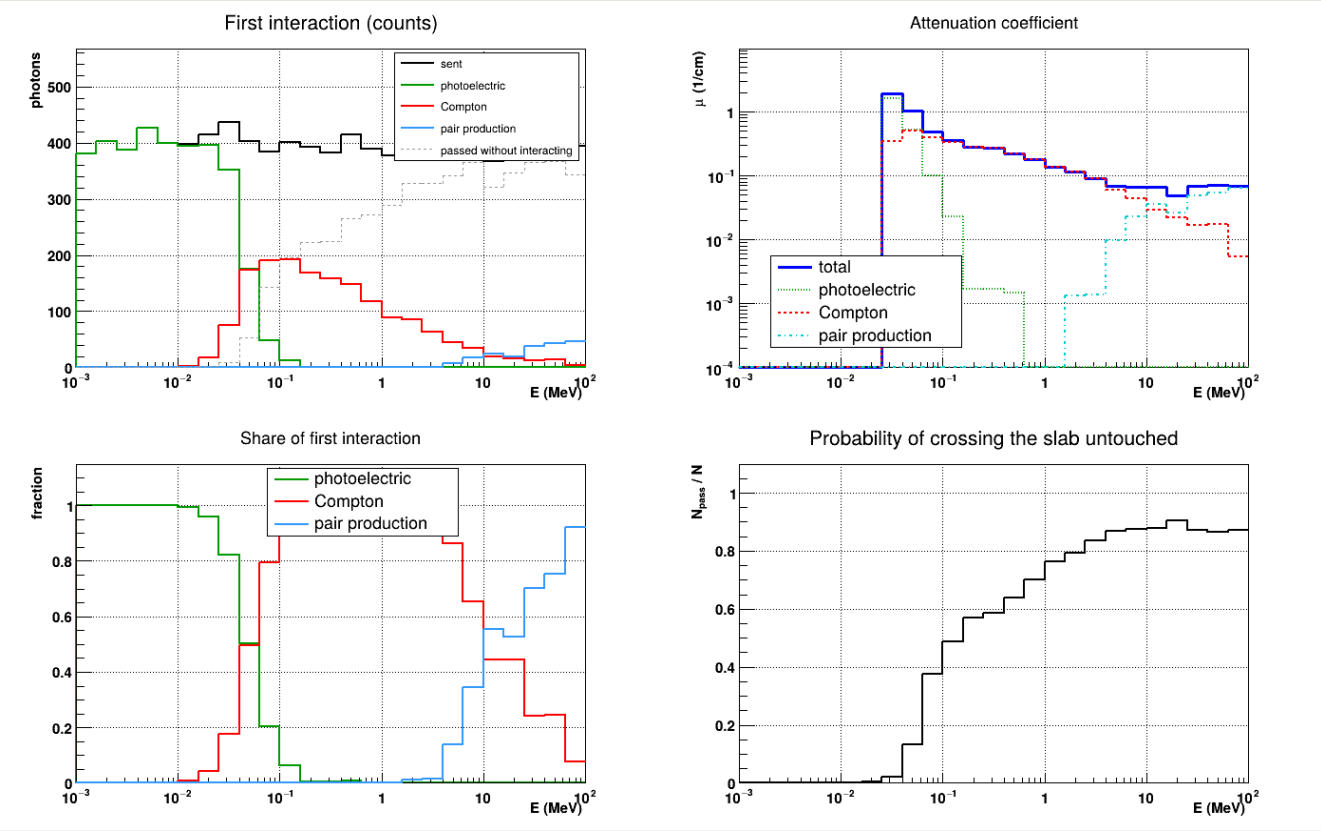
N: sent, N_{pass}: passed, n_i: photons whose first interaction is i · x: thickness (cm)

Three commands

```
cd ~/G4DetectorDemo-build
./demo gamma_scan_Al.mac
root -l 'draw_gamma.C("gscan_Al.root")'
```

enter the build directory
send the 10 000 photons; writes gscan_Al.root
draw the histograms with the ROOT macro

Result: which process dominates in aluminium?



Measured first-interaction fractions (2 cm Al)

E	Photo	Compton	Pair	Passed
1–10 keV	100 %	0 %	0 %	0 %
32 keV	82 %	18 %	0 %	2 %
50 keV	50 %	50 %	0 %	13 %
1 MeV	0 %	100 %	0 %	70 %
12.6 MeV	0 %	44 %	56 %	88 %
79 MeV	0 %	8 %	92 %	87 %

- **Photoelectric ≈ Compton:** ≈ 50 keV
- **Compton ≈ pair production:** ≈ 12 MeV
- Z = 13: the boundaries fall between water (Z ≈ 7.4) and germanium (Z = 32) of slide 27, as expected.

Homework: replace G4_Al by G4_Pb in the macro; the photoelectric region widens and the pair-production boundary drops to about 5 MeV.

Whatever its type, every detector can be described by a common set of properties that tell us what it can measure and how well.

S	$Q \propto E$	$\Delta E/E$	t	ε	τ
Sensitivity	**Response**	**Resolution**	**Timing**	**Efficiency**	**Dead time**
Does it give a usable signal for this radiation and energy?	How does the signal relate to the deposited energy?	How well are two nearby energies separated?	How fast does the signal form?	What fraction of the radiation is recorded?	How long is it blind after an event?

> **Figures of merit:** gas, scintillation and semiconductor detectors are all compared with these numbers.

Sensitivity and detector response

Sensitivity depends on:

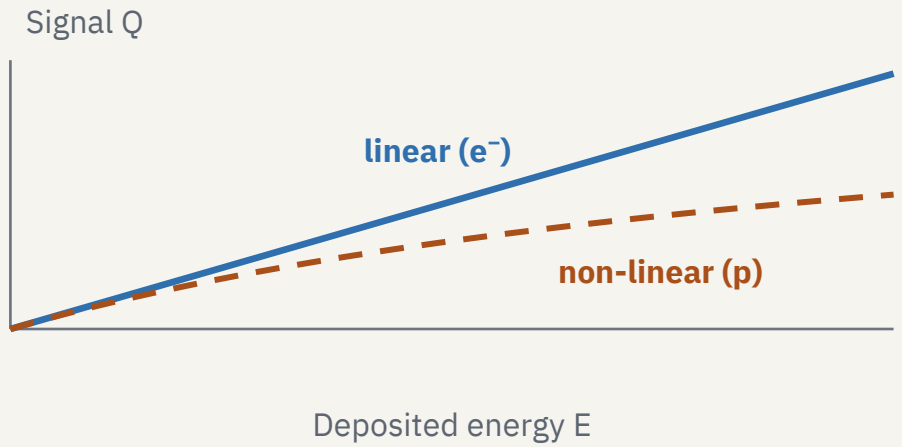
1. the cross section for ionizing reactions in the material
2. the detector mass
3. the inherent noise of the detector and its electronics
4. the protective material around the sensitive volume

> **Examples:** neutral particles need a large mass (neutrinos: tons)

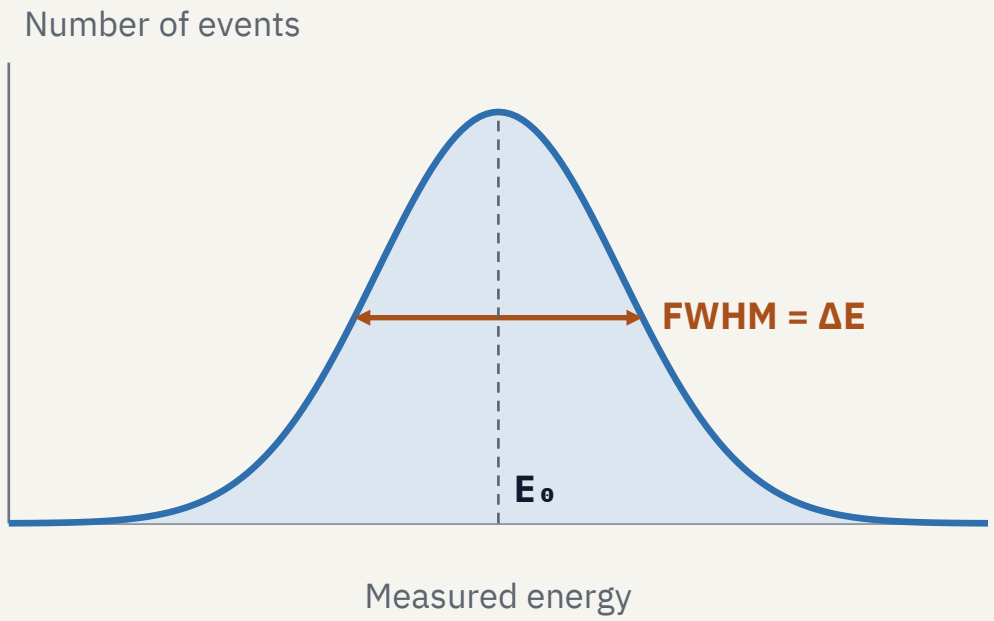
Response

Signal = a current pulse; its charge **Q** (or pulse height, if the shape is fixed) reflects the deposited energy: $Q \propto E$

Linearity depends on the particle: an organic scintillator is linear for e^- but not for p.



Energy resolution



FWHM: the width of the peak at half its height (ΔE).

Resolution: R = ΔE / E₀, usually in percent; the smaller R, the better the resolution

Why is there a width? The deposited energy E produces J charge carriers (ions or electron–hole pairs). J differs slightly from event to event: fluctuation σ = √J.

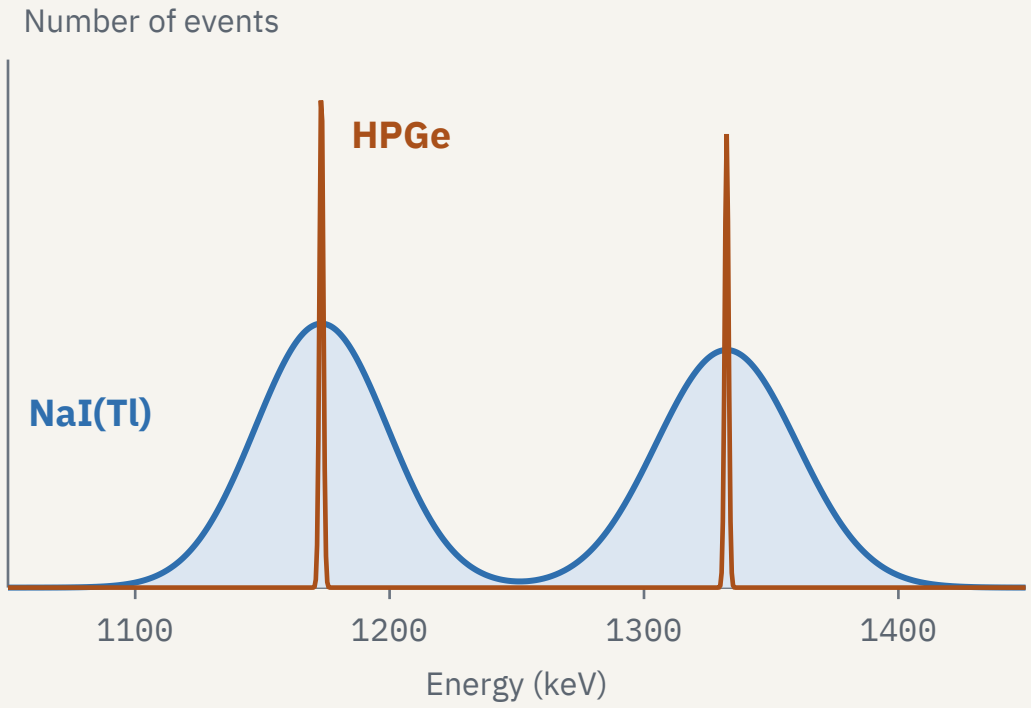
$$J = E / w, \quad R = 2.35 / \sqrt{J}$$

w: mean energy needed to create one carrier
· 2.35: for a Gaussian, FWHM = 2.35 σ

Example: Ge, 662 keV photon (w = 2.96 eV)

1) $J = 662\,000 / 2.96 \approx 224\,000$
2) $R = 2.35 / \sqrt{224\,000} = 2.35 / 473 \approx 0.50\ \%$
3) $\Delta E = 0.0050 \times 662\ \text{keV} \approx 3.3\ \text{keV}$

Resolution in practice: NaI and HPGe



The two γ lines of ^{60}Co (1173 and 1333 keV). HPGe: high-purity germanium.

Fano factor F

F = actual fluctuation / Poisson fluctuation. $\sigma^2 = F \cdot J$

Why F < 1? The particle deposits all its energy, so the total is fixed. If one ionization takes more energy, less is left for the others. The carriers cannot fluctuate independently, so the fluctuation shrinks.

$$R = 2.35 \cdot \sqrt{(F / J)}$$

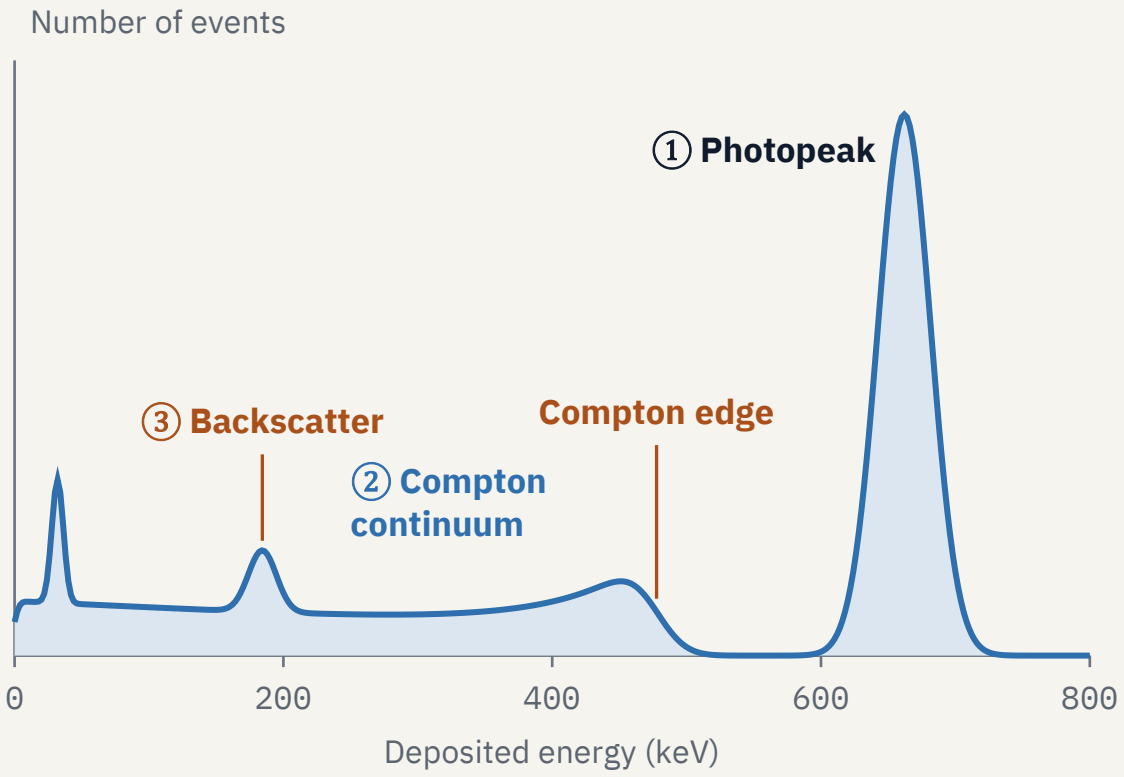
Ge, Si: F ≈ 0.1 · gas: 0.05–0.2 · scintillator: F ≈ 1

Example: 662 keV

Ge: $0.50\ \% \times \sqrt{0.13} \approx 0.18\ \% = 1.2$ keV

NaI: $\approx 7\ \% = 46$ keV (≈ 40 times wider)

Response function



^{137}Cs (662 keV) and a NaI detector, schematic

Response function: the spectrum a monoenergetic source produces in the detector.

① Photopeak: the photon deposits all its energy (662 keV).

② Compton continuum: the photon scatters and escapes; only the electron's energy is measured.

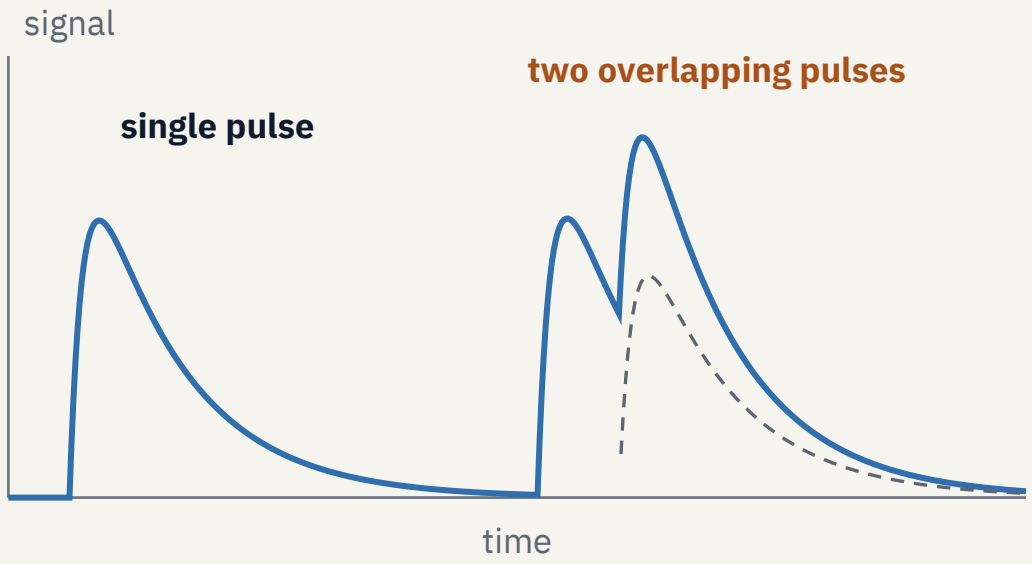
③ Backscatter: a photon scattered back from the surroundings enters the detector.

Compton edge: the maximum electron energy

$$T_{max} = E \cdot 2\varepsilon / (1 + 2\varepsilon), \quad \varepsilon = E / m_e c^2$$

1) $\varepsilon = 662 / 511 = 1.30$
2) $T_{max} = 662 \times 2.59 / 3.59 \approx 478$ keV
3) Backscatter: $662 - 478 = 184$ keV

Response time and pile-up



Response time: the time for the signal to form after the radiation arrives.

Pulse duration: how long the signal lasts. A second event arriving meanwhile piles up on it.

Detector	Typical time
Plastic scintillator	2–3 ns
NaI(Tl)	230 ns
Germanium (HPGe)	0.1–1 µs
Geiger–Müller tube	≈ 100 µs

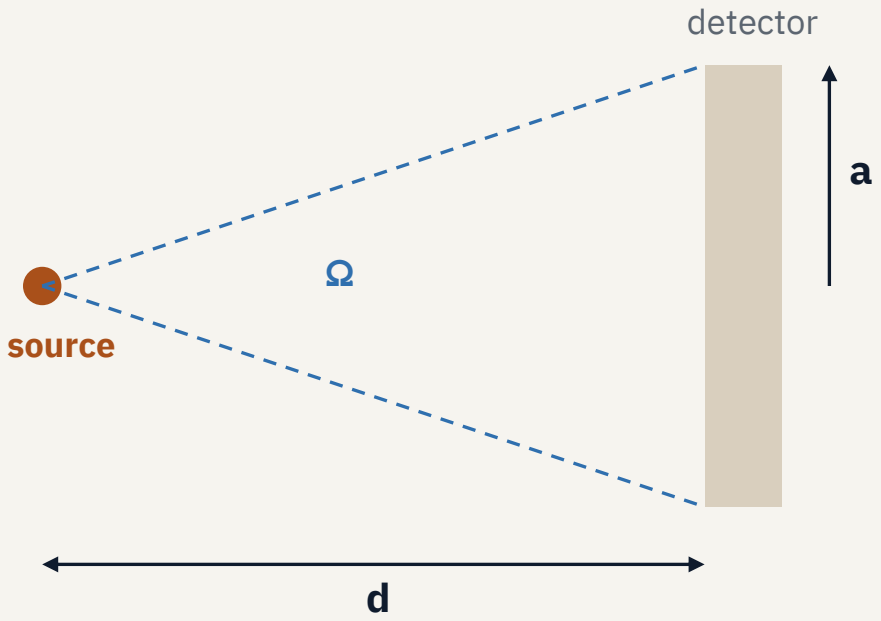
Example: pulse duration 1 µs, 100 000 events/s

Probability that another event arrives during a pulse ≈ count rate × pulse duration:

$$100\,000\ /\mathrm{s} \times 0.000001\ \mathrm{s} = 0.1$$

About 10 % of the events pile up.

Detection efficiency



$$\Omega = 2\pi \cdot \left(1 - d / \sqrt{d^2 + a^2}\right)$$

Ω: solid angle (sr), full sphere 4π · d: distance · a: detector radius

Geometric efficiency ε_{geo}: fraction heading to the detector, Ω / 4π

Intrinsic efficiency ε_{int}: fraction of the hits that is recorded

Total efficiency: $\varepsilon_{tot} = \varepsilon_{geo} \cdot \varepsilon_{int}$

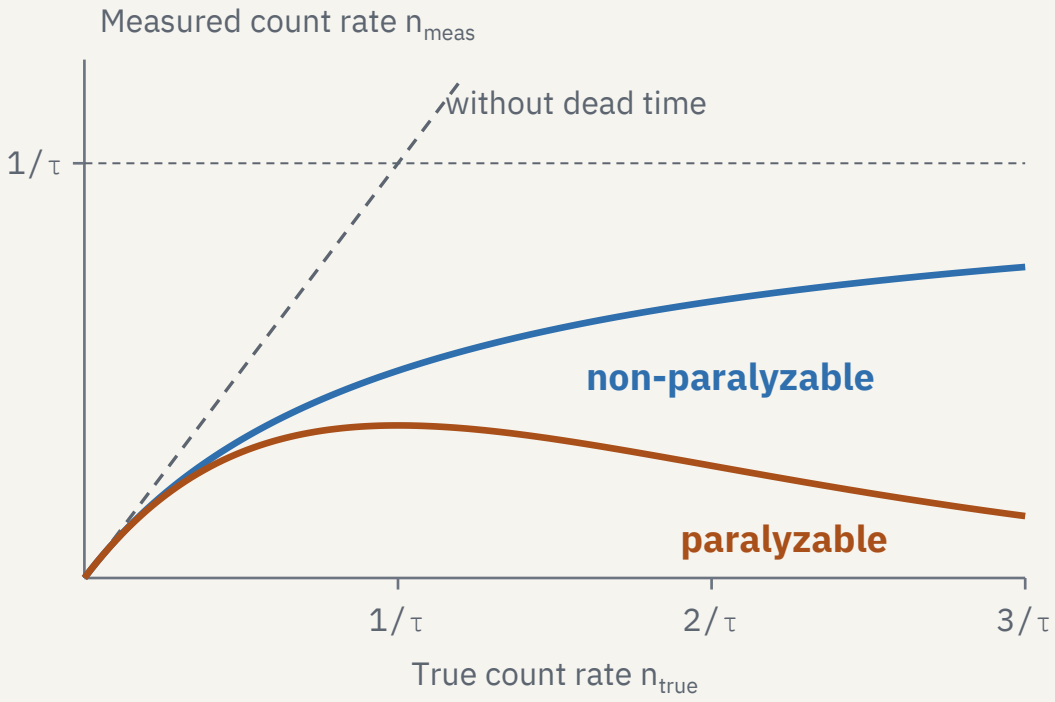
Example: ^{137}Cs, d = 10 cm, a = 2.5 cm, NaI

1) $\Omega = 2\pi\,(1 - 10/10.31) = 0.188$ sr
2) $\varepsilon_{geo} = 0.188 / 12.57 \approx 1.5\ \%$
3) $\varepsilon_{int} = 1 - e^{-0.28 \times 7.6} \approx 88\ \%$
4) $\varepsilon_{tot} = 0.015 \times 0.88 \approx 1.3\ \%$

In step 3, $\mu = 0.28\ \text{cm}^{-1}$ is the attenuation coefficient of NaI at 662 keV; 7.6 cm is the crystal thickness.

Of 1 million decays, about 13 000 are recorded.

Dead time



The more events arrive per second, the larger the fraction that is lost.

τ (dead time): the time after an event during which the detector cannot record a new one

n_{true}: the number of events actually reaching the detector per second

n_{meas}: the number of events the detector records per second

Non-paralyzable model: events in the dead time are lost.

$$n_{true} = n_{meas} / (1 - n_{meas} \cdot \tau)$$

Paralyzable model: every new event extends the dead time.

Example: τ = 10 µs, n_{meas} = 20 000 /s

1) $n_{meas} \cdot \tau = 20\,000 \times 0.00001 = 0.2$
2) $n_{true} = 20\,000 / 0.8 = 25\,000$ /s

Dead 20 % of the time; 20 % of the events are lost.

CLOSING

Summary and resources

Sources

α, β, γ, EC: known energies and branching ratios.

Radiation in matter

Bethe–Bloch and the Bragg peak, Cherenkov, radiation length, photon attenuation.

Detectors

Response, resolution (Fano), response function, efficiency and dead time.

Next step: these are exactly the processes a Geant4 physics list implements.

Further reading and data

- W. R. Leo — Techniques for Nuclear and Particle Physics Experiments, Ch. 1, 2, 5
- G. F. Knoll — Radiation Detection and Measurement
- NIST XCOM (photons), ESTAR / PSTAR / ASTAR (e, p, α)
- IAEA Live Chart of Nuclides (decay data)